

Simple Alpine VPS Setup Script

What is This Script?

- It's a **setup script** written in `sh` (POSIX shell)
- Used on a **VPS (Virtual Private Server)** running **Alpine Linux**
- Automates:
 - User creation
 - Security settings
 - Docker installation
 - Development tools setup

What is Alpine Linux?

- A lightweight, security-oriented Linux distribution
- Great for VPS and Docker
- Has small package manager: `apk`

Password Setup

```
PASSWORD=YOUR_PASSWORD
```

- You set this before running
- Used for the `deploy` user
- Must be **at least 8 characters**

Create User and Set Password

```
adduser -D deploy
```

- Creates a user named `deploy`

```
echo "deploy:$PASSWORD" | chpasswd
```

- Sets the password for that user

Sudo (Admin Rights)

```
apk add sudo  
addgroup deploy wheel
```

- `deploy` user is added to `wheel` group
- `sudo` lets user run admin commands

UFW Firewall Setup

```
apk add ufw
ufw default deny incoming
ufw allow 22 80 443 9000 3000 8080
ufw enable
```

- Blocks all incoming by default
- Allows specific ports: SSH, HTTP, HTTPS, etc.

Secure SSH

```
sed -i 's/#*PermitRootLogin.*/PermitRootLogin no/' /etc/ssh/sshd_config
```

- Prevents root from logging in via SSH
- SSH server is restarted after changes

Docker Installation

```
apk add docker docker-cli docker-compose  
rc-update add docker boot  
service docker start
```

- Installs and starts Docker
- Adds `deploy` to `docker` group

Install Development Tools

```
apk add git curl wget vim build-base zip unzip ...
```

- Basic developer tools
- Useful for editing, compiling, downloading

Fail2Ban for Brute Force Protection

```
apk add fail2ban
```

- Monitors logins and blocks attackers
- Configured to read `/var/log/auth.log` or `/var/log/messages`



Monitoring Scripts

- `/home/deploy/disk-check.sh`
 - Alerts if disk is almost full
- `/home/deploy/system-info.sh`
 - Shows system info and Docker status

Environment Config File

/home/deploy/.env

```
DEPLOY_USER=deploy  
ALLOWED_PORTS=22 80 443 ...  
DOCKER_INSTALLED=true
```

- Contains key setup info

✓ After Running the Script

- Login as `deploy` user with password
- Run: `sudo docker run hello-world`
- Use monitoring scripts
- Check logs: `/var/log/deploy-setup/setup.log`



Summary

This script helps you:

- Quickly set up a secure Alpine VPS
- Install Docker & dev tools
- Create a new deploy user
- Harden SSH & firewall settings

Try It Yourself!

1. Launch Alpine VPS (e.g. on Vultr)
2. Set your password:

```
export PASSWORD="YourStrongPassword"
```

3. Run the script:

```
sh setup.sh
```


🌟 Questions?

Let's experiment, break things, and learn together! 🚀